Mission Europa

An eTwinning project to test, improve and create physics games

Short overview | *October 2026 to April 2027*

Mission Europa is an eTwinning project for students aged 15–18, running from October 2026 to April 2027. Students from up to six schools work as international crews: they test space-themed physics games, decide how to improve them and design two new games for other students to play. Every design decision has to be defended with physics, and every school sees its ideas in the finished games.

We start with two games from the Mission ATLAS collection. Crews play them, report what worked and what confused them, and rank the changes they want; patch notes credit every change to the crew that proposed it. Then mixed international studios design two new games. For each one, the Greek team builds two playable versions from the studios' briefs, new players in every school test both versions blind, and the version they prefer goes forward with the best feature of the other. In April, every school launches the finished games to new players.

At a glance

Students	Aged 15–18
Schools	Up to six, from different countries
Dates	5 October 2026 to 30 April 2027, starting in World Space Week
Class time	12 lessons (core track) or 20 lessons (full track); at most one lesson a week, about 45 minutes each
Language	English in TwinSpace; students can prepare in their own language
Live meetings	None required; three short optional ones
What you need	A browser on a computer or tablet. Free games, no game accounts, no personal data, no programming
Join by	30 September 2026; schools can still join until 16 October (see How to join)

Try the starting games · all Mission ATLAS games: physicsplay.github.io/Astrojourneys



Asteroid Deflection — The 31.5 Rule

How small a nudge, given years in advance, moves an asteroid off a collision course.

Play in your browser



Comet Lander — Rosetta's Philae

Predict a landing in gravity ten thousand times weaker than Earth's, then face Philae's real bounce.

Play in your browser

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Detailed project description | *October 2026 to April 2027*

Why Mission Europa

Students play games every day, but few get to shape one that is played in other countries. Mission Europa gives them a real audience, real decisions and a reason to use physics when they make and defend those decisions. The physics involved (motion, free fall, gravitation, momentum and energy) belongs to the upper-secondary curriculum in every partner country, so the project runs inside physics lessons.

The project follows the eTwinning theme for 2026–2028, Future-Ready Schools, especially its focus on pupil voice and agency and on digital and AI literacy. It also runs alongside real missions: ESA's Hera arrives at the Didymos system, where NASA's DART struck the small asteroid Dimorphos in 2022, while the crews improve Asteroid Deflection, and NASA's Europa Clipper swings past Earth on its way to Jupiter's moon Europa while the studios design the first new game (see Key dates). *From Europe to Europa.*

How we work together

- **Crews.** Each school's students form a crew and choose call signs, which appear in credits and patch notes instead of real names. Crews design mission patches and post a 60-second introduction (avatar, voice or video, within each school's consent rules).
- **Twin crews.** Each crew is paired with a twin crew in another country for the whole project (a group of three if the number of schools is odd). Twins reply to each other's findings first and review each other's design briefs.
- **International studios.** For each new game, the schools form two studios of two or three schools each, and each studio designs one version. Studios are reshuffled for the second game, so every school works directly with most of the others.
- **Roles by interest.** Students choose a role and can change it between phases: physics officer, narrative designer, artist or sound designer, tester (bugs and accessibility), translator or reporter. Two or three reporters per school become TwinSpace pupil administrators and keep the project pages up to date.
- **Mission Control.** The Greek team meets twice a week: it checks the numbers, analyses the mission reports and writes the patch notes, and builds the games outside lesson time. The designs belong to the studios. A partner school may help with programming if it wishes; it is never required.
- **English without barriers.** Students discuss ideas in their own language first, then contribute in English through short texts, voice notes, annotated screenshots, sketches or clips. Sentence starters are provided, including Project Zero's Claim–Support–Question routine for explaining choices.

> **A Mission Europa lesson (45 minutes)**
>
> 5 min briefing: this week's mission and the twin crew's latest post · 20 min play one stage in groups of two or three · 10 min feedback card and mission report · 10 min post to TwinSpace and reply to the twin crew. Design lessons follow the same shape, with the design brief in place of the game.

Testing and improving the existing games

We begin with Asteroid Deflection and Comet Lander. Before the first playtest, each game gets a tester mode that shows the version number, lets a group jump straight to one stage so that a test fits into a lesson, and resets progress on shared devices.

At the end of a session, each group copies an anonymous one-line mission report from the game (version, stage, device type, attempts, error, hints used, time and two or three ratings from 1 to 5) and pastes it into the TwinSpace forum. The game itself stores nothing and collects no personal data.

Groups also post feedback cards. A card names the game and version, describes the incident behind the suggestion (for example a confusing graph, a hint that reveals too much or a time limit that leaves no room to explain reasoning), says why it matters and proposes a change. The Greek team marks each proposal S, M or L for development effort, and the crews rank the proposals in a TwinSpace poll. When crews propose two good alternatives, both can be built and compared.

Patch notes appear within 7–10 days and credit the crew behind each change. Every version keeps its own stable link, so crews can compare the old and new versions and decide whether a change solved the original problem or should be reverted.

Designing and selecting new games

> **Mission rules for every new game**
>
> 1. One physics relationship per game.
> 2. Players commit their number before the simulation runs.
> 3. The game never shows the answer before the player commits.
> 4. Four stages: Watch → Predict → Inverse → Crisis. A bronze medal unlocks the next stage.
>
> *Everything else is up to the studios: setting, story, characters, what the player controls, how the game helps and responds, and how it looks and sounds.*

Studios choose the physics from a short menu of relationships that teachers have already checked, for example the inverse-square law (why Europa Clipper needs such large solar arrays), Kepler's third law (the 1:2:4 rhythm of Io, Europa and Ganymede) or momentum transfer in a kinetic impact (what Hera is investigating at Dimorphos). Teachers can add options before the vote.

Each studio then writes a one-page design brief with four boxes, one for each stage: what the player watches, predicts, works out in reverse and faces in the crisis. The brief also describes the setting, the player's role, the help available and how success is recognised. Twin crews review the brief, and teachers check the physics and feasibility before it is locked. Crews can also contribute content that goes straight into a game without programming: crisis events, hints, briefings, character lines, debrief facts and translations.

The Greek team builds both versions of a game as one file with an A/B switch, on the same engine and verified scenarios, so the versions differ only in the studios' design choices. The games are built with AI-assisted programming: students see how a brief becomes a working game, teachers check scientific accuracy and basic functionality before every comparison, and crews hunt for any remaining physics errors.

Each school then runs a blind playtest with new players from another class: each new player plays one stage of version A and one of version B (about 15 minutes each; half start with B), while project students observe and take notes. Each school's result counts once, whatever the size of the class. The version preferred in most schools goes forward; a tie is decided by the learning data (smaller errors, fewer hints). The crews then vote in a ranked-choice poll on which features of the other version to carry over and which improvements come next, so no studio's work is lost.

At the end of January, the teachers review progress. If time is short, the second game becomes a single version to which each school contributes one level or crisis pack.

Recognition and challenges

- **Credits and certificates.** Every game has a credits screen listing the crews and studios by call sign, a Hall of Fame page in TwinSpace shows who changed what, and each student receives a certificate listing their role and contributions.
- **Challenges.** A bug bounty with a crew leaderboard credits confirmed bugs and physics errors, and a blooper reel collects the most spectacular crash landings, Philae's real bounce included.

Shared timetable

The windows are shared; each school chooses its own lesson dates inside them and the windows are adjusted at the kick-off where holidays require it. Every school takes part in every phase; the core track uses fewer lessons per phase and the full track uses all 20.

When	Phase and main activities	Core	Full
5–30 Oct	**1. Launch and first playtest** Kick-off in World Space Week: crews, call signs, mission patches, 60-second introductions. Play both games in tester mode; post mission reports and feedback cards; twin crews reply; rank the changes.	2	3
2–20 Nov	**2. Patch and retest** Compare old and new versions and decide what to keep. Meet your studio, read the mission rules and vote on the physics for Game 1. Follow Hera's arrival at Didymos.	1	2
23 Nov – 29 Jan	**3. Game 1** Studios write the brief; twin-crew review and physics check; brief locked on 15 Dec. The Greek team builds versions A and B over the winter break. January: blind playtest with new players, decision, carry-over vote and next improvements.	4	6
1 Feb – 19 Mar	**4. Game 2** Scope confirmed at the January checkpoint. Reshuffled studios write the brief, locked on 19 Feb. Retest the improved Game 1 while Game 2 is built. Blind playtest, decision and carry-over vote by 19 Mar.	3	5
22 Mar – 16 Apr	**5. Final polish** Final test round on the schools' devices: instructions, difficulty, feedback and accessibility. Last corrections, translations and credits. Planned around the Easter holidays.	1	2
19–30 Apr	**6. Launch and review** Launch Week, 19–23 Apr: each school presents the games to new players, with an optional live link. Review and reflection by 30 Apr.	1	2
Total	**Every school takes part in every phase; at most one lesson a week**	**12**	**20**

Key dates

By 2 Oct 2026	Teachers, no lessons: TwinSpace pupil accounts, media consent, school holidays in the shared calendar
4–10 Oct 2026	World Space Week: project kick-off
November 2026	Hera due at Didymos
3 Dec 2026	Europa Clipper flies past Earth
End of Jan 2027	Progress checkpoint; scope of Game 2 confirmed
28 Mar 2027	Western Easter; Phase 5 is planned around the Easter holidays
19–23 Apr 2027	Launch Week. The Greek school is closed for Orthodox Easter from 26 April to 9 May, so the Greek team completes its launch and review by 23 April.

Participation and practical arrangements

Each partner teacher organises the local crew, supports short contributions in English, helps check the physics and keeps to the agreed windows. Blind playtests need new players, such as another class, for part of a lesson; Launch Week needs an audience.

Testing needs a computer or tablet with a browser; devices can be shared within groups. The games are self-contained HTML files with no external scripts, no backend, no accounts and no personal data, checked on computers and tablets before each release. Every version is published at one project address, missioneuropa.eu, with a stable link and a QR code, and can be downloaded for offline play.

TwinSpace holds a Start here page with the mission rules and templates, one forum thread per game and version, the Decision Log (suggestion, decision, result), the Patch Notes, the Hall of Fame and the polls.

No live meetings are required. Three optional online meetings of 20–30 minutes in TwinSpace (the kick-off, the A/B reveal in January and Launch Week) can also run as pairs of schools, with a short summary posted for everyone.

Students communicate only in TwinSpace, with accounts managed by their teachers, and appear in credits under their call signs. Photos, voices and faces are shared only within each school's consent rules. Student contributions (art, text, sound, translations) are published under a CC BY licence with credit to the crew; third-party images are credited according to their licences.

What partner schools get

- **Ready-made templates:** feedback card, design brief, lesson pattern and sentence starters.
- **A predictable workload:** core or full track, one lesson a week at most, no programming, no required meetings.
- **Support for the National Quality Label:** shared documentation and evidence for your application.
- **Recognition for students:** credits in the games, certificates and a real audience across Europe.

Planned outcomes: improved versions of the two starting games, two new games designed by international studios, crew translations and a public record of how feedback shaped every game. Students practise applying physics, testing predictions, explaining evidence and revising designs.

How to join

Please contact the project coordinator, Anastasios (Tasos) Gkinis, physics teacher, 2nd General Lyceum of Moschato (Athens, Greece), through the European School Education Platform. Reply by 30 September 2026 if you can; schools can still join until 16 October, during Phase 1. Tell us your school, country and subject, the age and number of your students, core or full track, and the weeks your school is closed.